Fiscal Year:	FY 2016	**Task Last Updated:**	FY 05/24/2016
PI Name:	Blancaflor, Elison B. Ph.D.		
Project Title:	Utilizing the Advanced Biological Research System (ABRS) on the International Space Station (ISS) to Uncover Microgravity's Impact on Root Development and Cell Wall Architecture		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	elison.b.blancaflor@nasa.gov	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	321-501-5368
Organization Name:	NASA Kennedy Space Center		
PI Address 1:	Utilization and Life Science Office		
PI Address 2:	Exploration and Research Technology Programs		
PI Web Page:			
City:	Kennedy Space Center	**State:**	FL
Zip Code:	32955	**Congressional District:**	8
Comments:	NOTE: Dr. Blancaflor moved to NASA Kennedy Space Center in spring 2021; as a PI, his prior grants were for work at the Noble Foundation in Oklahoma.		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	07/25/2012	**End Date:**	07/24/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 7/24/2017 per PI and NSSC information (Ed., 7/13/16) NOTE: Extended to 7/24/2016 per PI and NSSC information (Ed., 5/27/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nakashima, Jin (Noble Foundation) Tang, Yuhong (Noble Foundation)		
Grant/Contract No.:	NNX12AM94G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of this proposal is to uncover the underlying molecular mechanisms by which microgravity impacts cell wall architecture in the model plant Arabidopsis thaliana using the Advanced Biological Research System (ABRS) housed on the International Space Station (ISS). This proposal builds on a robust dataset from our rapid turnaround spaceflight experiments, which utilized the Biological Research in Canisters (BRIC16) hardware on the STS-131 mission. Through experiments with the BRIC hardware, we discovered that spaceflight had a profound influence on the expression of genes involved in regulating cell wall architecture. This led us to hypothesize that plants respond to microgravity in large part through the transcriptional reprogramming of genes that control cell wall remodeling. Because of the prominent root developmental phenotypes (i.e., primary root skewing and inhibition of root hair growth) that we observed in our BRIC experiments, it appears that microgravity-dependent changes in cell wall-related transcripts are predominantly manifested in roots. Furthermore, because primary root and root cell wall ultrastructural defects were most pronounced in a mutant to a vegetative actin isoform (act2), we hypothesize that microgravity predominantly affects genes that lie within signaling networks linking the actin cytoskeleton with the plant extracellular matrix. Given our successful experiments with BRIC, we are in an excellent position to effectively utilize the unique capabilities of the ABRS hardware on the ISS to test the aforementioned hypotheses. We will use a new algorithm that we developed from our BRIC16 experiments to more rigorously quantify root growth behavior of wild-type and a higher order vegetative actin mutant (act2 act8 double mutant) (aim 1). This will clarify the extent of actin involvement in specifying root orientation and cell wall remodeling in space. Through the use of monoclonal antibodies and immunocytochemistry, we will also determine how the distribution and abundance of specific cell wall components change in microgravity, and correlate such changes with root developmental responses and modifications in gene expression (aim 2). Finally, we will employ more sensitive deep sequencing technologies (e.g., RNA-Seq) to discover novel microgravity-induced transcriptional changes in plants (aim 3). Knowledge from our spaceflight studies will then be translated into corresponding gene functional studies (e.g., mutant analysis and overexpression), which we expect will lead to the discovery of new players that modulate root development on Earth and in microgravity. The proposed work is therefore relevant to this NASA-NRA because it will address fundamental questions on how gravity modulates the plant's internal cytoarchitecture and the molecular processes that govern the development of its cell walls. This will be significant for NASA's mission as extrapolated in the priorities of the Fundamental Space Biology Plan for 2010-2020 because it will provide us with basic knowledge to guide genetic engineering efforts to generate plants that are better adapted to the weightless environment of spaceflight. This in turn will be beneficial for the design of large plant habitats on the ISS and future space colonies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide a deeper understanding of how microgravity influences plant development. The focus of this research will be on the development of roots and how they modify the expression of their genes to adapt to the microgravity environment of spaceflight. This research will address one of high priority recommendations of the National Research Council's 2011 decadal survey report, which is to contribute significantly to building a robust spaceflight program to analyze plant physiological responses to stimuli encountered in microgravity. Research areas depicted in this proposal, namely root development and cell wall biology, are significant for NASA's goal of developing large plant habitats on the International Space Station (ISS) and future space colonies. We expect that information gained from basic studies of root growth in microgravity will guide the development of plant varieties better equipped to optimize nutrient and water acquisition, and waste recycling in enclosed regenerative life support systems. Furthermore, because upright plants evolved under the 1-g conditions of Earth in large part through the mechanical strength of its cell walls, this research will provide a deeper mechanistic understanding of the genetic networks that specify how cell walls are constructed in microgravity. In space, such knowledge will enable technologies for designing plants that can sustain optimal growth and stature to overcome the adverse impact that microgravity has on cell wall development. With regard to Earth benefits, understanding of molecular pathways that govern root growth and cell wall development, will benefit agricultural practices and bioenergy research on Earth.
Task Progress:	We have made substantial progress with this project during this reporting period. We can attribute our success to the fact that in-orbit operations were completed in early 2015 allowing us to process our spaceflight samples during this reporting period. Highlights of our progress include the following: 1). We verified that directional root skewing occurs in microgravity. These results are consistent with previous spaceflight studies on a different set of hardware (i.e., BRIC). For our current experiments, we used the VEGGIE hardware and seedlings were grown under continuous light. However, similar skewing responses were observed in the BRIC hardware for seedlings grown in continuous dark. 2). Assembly and mapping of the sequence reads from RNA-Seq experiments were completed during this reporting period. In summary, total sequence reads from the samples ranged from 75- 120 million with more than 95% mapping to the Arabidopsis genome (TAIR version 10). Basic statistical analyses indicate high quality reads from our transcriptomic efforts. We found that genes differentially regulated by microgravity could be classified into six gene ontology (GO) clusters based on biological function. Interestingly, genes in the three GO clusters overrepresented were involved in root hair growth and oxidative stress (e.g., peroxidases). It is noteworthy that similar classes of genes were shown to be repressed in microgravity from dark-grown Arabidopsis seedlings on our previous BRIC 16 experiments. Our most recent transcriptomics data continue to point to the spaceflight environment as a major trigger for oxidative stress responses in plants. This indicates that developing the ideal plant ideotype for future space habitats could be patterned from plant culitvars that have been developed for marginal agricultural lands including crops tolerant to low oxygen (e.g., hypoxia) and other stressors (drought, insect resistance) as well as those with more robust cell wall mechanical properties. 3). We demonstrated the feasibility of using glycome profiling to assess the impact of microgravity on cell wall composition. We obtained excellent results from a small amount of space flown tissue (i.e., we were able to extract cell walls from plant debris from columns used for RNA extraction). This allowed us to optimize the use of limited spaceflight material. In summary, our glycome profiling work indicate that microgravity triggers alterations in cell wall composition in part because of changes in expression of genes encoding cell wall remodeling proteins. Further, we demonstrated the feasibility of glycome profiling to identify specific components of the plant cell wall that are altered by spaceflight. Finally, immunolabeling using mABs of plant sections enabled us to identify cell/tissue types where microgravity-triggered cell wall composition occurred.

	4). Ground-based studies led to the discovery of a novel protein involved in actin-mediated post-Golgi trafficking. This novel protein was named HLB1 for hypersensitive to latrunculin B 1. Latrunculin B is a potent compound that disrupts the actin cytoskeleton. The HLB1 protein was discovered through a forward genetic screen of Arabidopsis seedlings grown on latrunculin B. Our results have been published in the high-impact journal Plant Cell because the work has led to the discovery of a protein complex that links vesicular trafficking in plants with activities of the cytoskeleton.
Bibliography Type:	Description: (Last Updated: 01/13/2026)
Abstracts for Journals and Proceedings	Nakashima J, Avci U, Pattathil S, Sparks JA, Hahn MG, Blancaflor EB. "Immunological approaches to evaluate cell walls in Arabidopsis seedlings grown in microgravity." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. Concurrent session 6: Plants #1. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Blancaflor EB. "Identification of a novel protein involved in actin-mediated directional root skewing from ground- and space-based plant experiments." Presented at Plant Biology 2015, Minneapolis, Minnesota, July 26-30, 2015. Space Biology Minisymposium at the annual meeting of the American Society of Plant Biologists. Plant Biology 2015, Minneapolis, Minnesota, July 26-30, 2015. , Jul-2015
Abstracts for Journals and Proceedings	Sparks JA, Kwon T, Nakashima J, Allen SN, Tang Y, Blancaflor EB. "Progress on processing of Arabidopsis thaliana samples from Advanced Plant Experiments in Space (APEX 03-1)." Presented at the 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. IP 32. , Nov-2015
Abstracts for Journals and Proceedings	Kwon T, Sparks JA, Nakashima J, Allen SN, Krom N, Tang Y, Blancaflor EB. "Transcription profiles of Arabidopsis thaliana grown in space reveal regulatory networks for the adaptive response to microgravity." Presented at the 2016 Meeting of the Southern Section of the American Society of Plant Biologists, Denton, Texas, April 2–4, 2016. 2016 Meeting of the Southern Section of the American Society of Plant Biologists, Denton, Texas, April 2–4, 2016. , Apr-2016
Abstracts for Journals and Proceedings	Nakashima J, Avci U, Pattathil S, Sparks JA, Hahn MG, Blancaflor EB. "Immunological approaches to evaluate cell walls in Arabidopsis seedlings grown in microgravity." Presented at the XIV Cell Wall Meeting, Chania, Crete, Greece, June 12-17, 2016. Concurrent Session 3: Cell biology and dynamics of plant cell wall. XIV Cell Wall Meeting, Chania, Crete, Greece, June 12-17, 2016. , Jun-2016
Articles in Peer-reviewed Journals	Sparks JA, Kwon T, Renna L, Liao F, Brandizzi F, Blancaflor EB. "HLB1 is a tetratricopeptide repeat domain-containing protein that operates at the intersection of the exocytic and endocytic pathways at the TGN/EE in Arabidopsis." Plant Cell. 2016 Mar;28(3):746-69. http://dx.doi.org/10.1105/tpc.15.00794 ; PubMed PMID: 26941089; PubMed Central PMCID: PMC4826010 , Mar-2016
Articles in Peer-reviewed Journals	Dyachok J, Paez-Garcia A, Yoo CM, Palanichelvam K, Blancaflor EB. "Fluorescence imaging of the cytoskeleton in plant roots." Methods Mol Biol. 2016;1365:139-53. (Cytoskeleton Methods and Protocols: Methods in Molecular Biology series 3rd edition) http://dx.doi.org/10.1007/978-1-4939-3124-8_7 ; PubMed PMID: 26498783 , Jan-2016